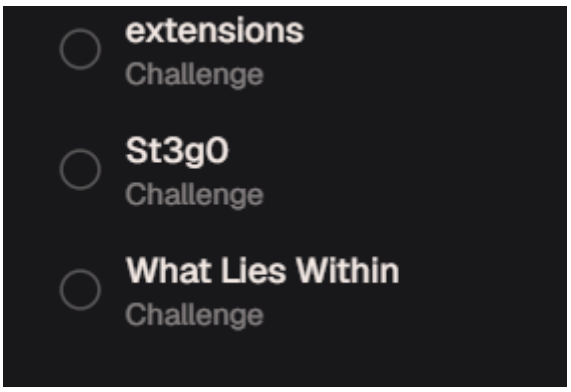
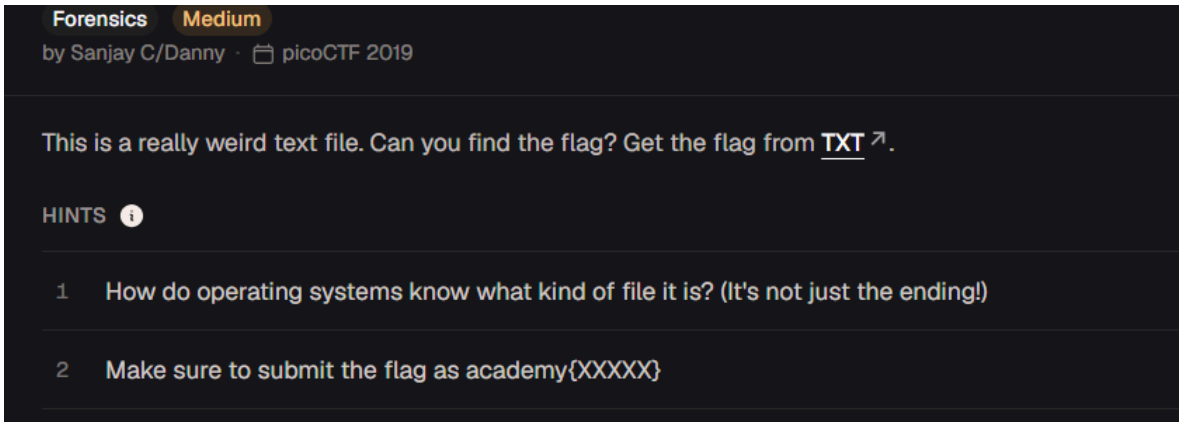
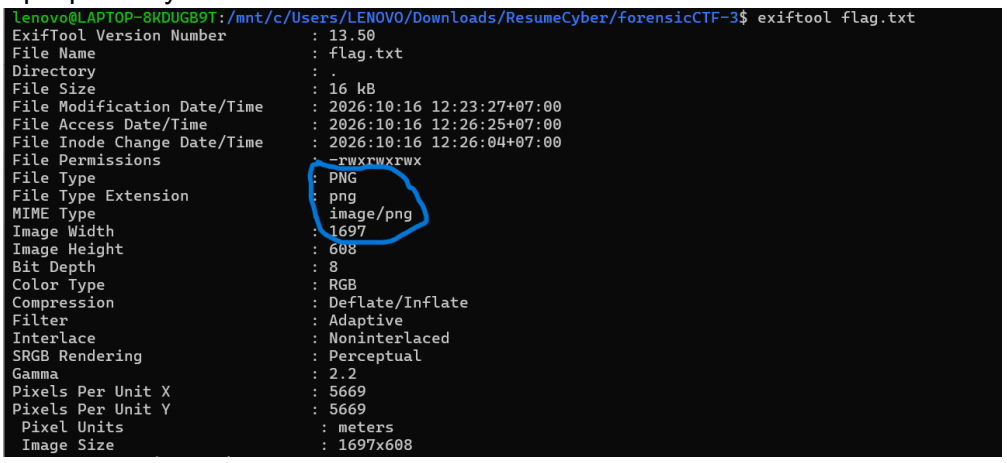


Challenge 1 : extensions

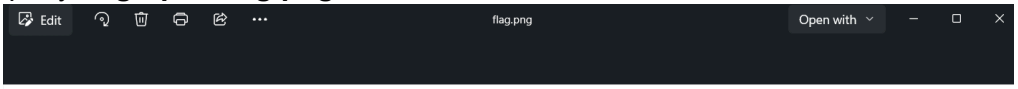


File Bernama : flag.txt

1. Kita lihat informasi filenya dan menemukan kejanggalan yakni file bernama flag.txt tapi tipe filenya PNG.



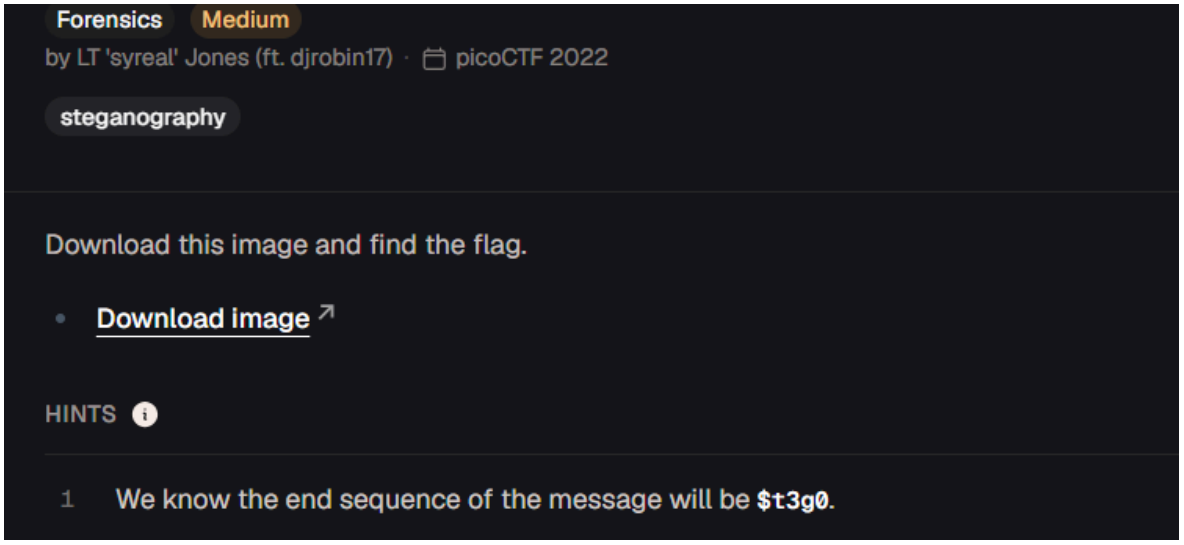
2. Karena kejanggalan itu kita harus mengubah filenya dari flag.txt menjadi flag.png dengan query **mv flag.txt flag.png**
3. Setelah menjadi PNG, kita buka file tersebut dengan library xdg-open. contoh query **xdg-open flag.png**



academy{now_you_know_about_extensions}

4. Ketemu flagnya yakni `academy{now_you_know_about_extensions}`

Challenge 2 : st3g0



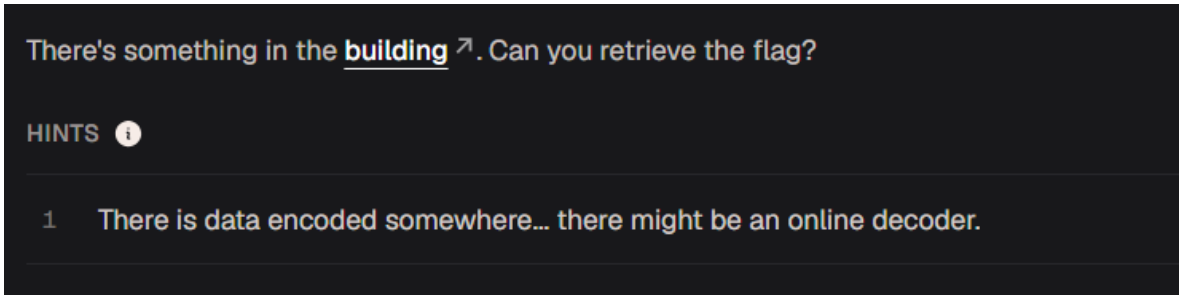
File bernama : academy.flag.png

1. Seperti biasa kita melakukan pengecekan file melalui **exiftool**.
2. Karena di soal ada ketentuan steganography, maka kita langsung next untuk cek steganography (skip **strings dan binwalk**).
3. Kita bakal menggunakan library zsteg (**zsteg academy.flag.png**)

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-3$ zsteg academy.flag.png
b1,rgb,lsb,xy       .. text: "academy{7h3r3_15_n0_5p00n_7e497237}$t3g0"
b1,rgba,msb,xy      .. text: "2TAF0RU6r"
b1,abgr,lsb,xy      .. text: "E2A5q4E%uSA"
b2,b,lsb,xy         .. text: "AAAPAAQTAAA"
b3,r,msb,xy         .. file: Applesoft BASIC program data, first line number 18
b4,r,lsb,xy         .. file: Targa image data - Map (16-257) 65536 x 4097 x 1 +257 +4369 - 1-bit alpha - right
b4,b,lsb,xy         .. file: Targa image data 272 x 17 x 16 +272 +272 - 1-bit alpha "\020\001\021\001\021\020\02
0\001\020\001"
```

4. Ketemu flagnya yakni `academy{7h3r3_15_n0_5p00n_7e497237}`

Challenge 3 : what lies within



File bernama : buildings.png

1. Sama seperti sebelumnya, kita lakukan exiftool terlebih dahulu.
2. Karena ga menemukan apapun kita lakukan strings dan binwalk

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-3$ exiftool buildings.png
ExifTool Version Number         : 13.50
File Name                       : buildings.png
Directory                       : .
File Size                       : 616 kB
File Modification Date/Time     : 2026:10:16 13:29:37+07:00
File Access Date/Time           : 2026:10:16 13:30:20+07:00
File Inode Change Date/Time     : 2026:10:16 13:30:22+07:00
File Permissions                : -rwxrwxrwx
File Type                       : PNG
File Type Extension             : png
MIME Type                       : image/png
Image Width                     : 657
Image Height                    : 438
Bit Depth                       : 8
Color Type                      : RGB with Alpha
Compression                     : Deflate/Inflate
Filter                          : Adaptive
Interlace                       : Noninterlaced
Image Size                      : 657x438
Megapixels                      : 0.288
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-3$ strings buildings.png | grep -i "academy"
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-3$ binwalk buildings.png

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
0             0x0             PNG image, 657 x 438, 8-bit/color RGBA, non-interlaced
41            0x29            Zlib compressed data, best compression
220405        0x35CF5         JBOOT STAG header, image id: 2, timestamp 0xE6FB789D, image size: 1990138519 bytes, image JBOOT checksum
: 0x1801, header JBOOT checksum: 0x986C
```

3. Selanjutnya kita coba ke zsteg dan yap kita menemukan flagnya

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-3$ zsteg buildings.png
b1,rgb,lsb,xy       .. text: "academy{h1d1ng_1n_th3_b1t5}"
b2,b,lsb,xy         .. text: "XuH}p#8Iy="
b3,abgr,msb,xy      .. text: "t@Wp-_tH_v\r"
b4,r,lsb,xy         .. text: "fdD\"\"\"\" "
b4,r,msb,xy         .. text: "%Q#gpSv0c05"
b4,g,lsb,xy         .. text: "fDfffDD\"\""
b4,g,msb,xy         .. text: "f\"fff\"\"DD"
b4,b,lsb,xy         .. text: "\"$BDDDDf"
b4,b,msb,xy         .. text: "wwBDDDfUU53w"
b4,rgb,msb,xy       .. text: "dUcv%F#A`"
b4,bgr,msb,xy       .. text: " V\"c7Ga4"
b4,abgr,msb,xy      .. text: "gOC_$_@o"
```

4. Ketemu flagnya yakni `academy{h1d1ng_1n_th3_b1t5}}`